MADAN MOHAN MALAVIYA UNIVERSITY OF TECHNOLOGY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2024-25) -- SEMESTER 1

COURSE: ENGINEERING PHYSICS (BMA-101)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: BMA-101

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) State the postulates of Special Theory of Relativity.

Q1. (b) Write physical significance of Maxwell four electromagnetic equations.

Q1. (c) What is de-Broglie hypothesis? Write formula for de-Broglie wavelength.

Q1. (d) Define Poynting vector and write its SI unit.

Q1. (e) What is population inversion in lasers? Why is it essential for lasing action?

Q1. (f) Differentiate between step-index and graded-index optical fibers.

Q1. (g) Explain the concept of length contraction in relativistic mechanics.

Q1. (h) What is Heisenberg Uncertainty Principle? State its mathematical inequality.

Q1. (i) Define numerical aperture and acceptance angle in optical fibers.

Q1. (j) State Brewster law of polarization with diagram.

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Derive Einstein mass-energy equivalence relation E = mc^2 from relativistic mechanics.

Q2. (b) Derive electromagnetic wave equations in free space and prove that light travels with speed c.

Q2. (c) Derive time-independent Schrodinger wave equation for a particle in a 1D box and find energy eigenvalues.

Q2. (d) Explain construction and working of He-Ne laser with energy level diagram.

Q2. (e) Explain optical fiber communication system with block diagram and dispersion mechanisms.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Derive Lorentz transformation equations for space and time coordinates. [10 Marks]

-- OR --

(b) Show that the circle x^2 + y^2 = a^2 appears as an ellipse to an observer moving with velocity v. [10 Marks]

Q4. (a) Derive continuity equation from Maxwell equations and explain displacement current. [10 Marks]

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(b) State and prove Poynting theorem for the conservation of electromagnetic energy. [10 Marks]

Q5. (a) Solve Schrodinger equation for a particle in a 1D infinite potential well and plot wavefunctions. [10 Marks]

-- OR --

(b) Explain physical interpretation of wavefunction psi and normalization condition. [10 Marks]

Q6. (a) Explain the working of Ruby laser with three-level diagram and output waveform. [10 Marks]

-- OR --

(b) Explain spontaneous emission, stimulated emission, and Einstein A and B coefficients. [10 Marks]

Q7. (a) Derive expression for numerical aperture (NA) and acceptance angle of an optical fiber. [10 Marks]

-- OR --

(b) Discuss attenuation mechanisms (absorption, scattering, bending losses) in optical fibers. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****